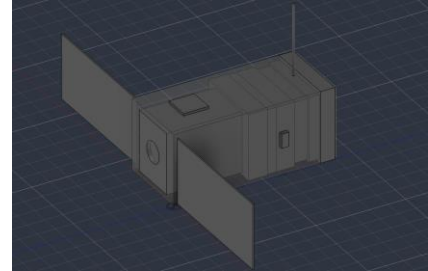


Chancellor Delk

AURORA-1 | 2U Earth-Observation CubeSat Preliminary Design

Overview

Independent digital spacecraft engineering project focused on the preliminary design of a 2U Earth-observation CubeSat in a 500 km circular LEO. The project integrates requirements development, subsystem trades, MATLAB analysis, power and mass budgeting, communications, ADCS, thermal modeling, and Autodesk Fusion CAD packaging.



Architecture

- 2U CubeSat, $\sim 100 \times 100 \times 227$ mm
- Gecko-class visible-light imaging payload
- UHF command/telemetry + S-band image downlink
- Deployable solar arrays
- Integrated ADCS with reaction-wheel control

Metric	Result
Orbital period	94.61 min
Eclipse duration	35.75 min
Average power	3.59 W
Solar target	7.50 W
Min. downlink rate	100 kbps
Pointing requirement	$\pm 2^\circ$
Simulated pointing error	0.382°

Major Design Decision

A 1U baseline was replaced with a 2U architecture after mass, volume, payload, ADCS, and power trades showed significantly better design margin.

Tools & Skills

MATLAB • Autodesk Fusion • Excel • Systems Engineering • Trade Studies • Power Budgeting • Link Budgeting • Thermal Modeling • ADCS Analysis • CAD Packaging

Status

Preliminary digital design baseline complete. Future work includes orbital lifetime, higher-fidelity thermal analysis, EPS refinement, and three-axis ADCS analysis.